

Lecture 4: Kalman Filter

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Introduction to Data Assimilation
2026 Spring



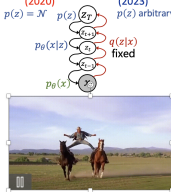
Diffusion Models for DA

Energy/Score-based
generative models
(2019)



Diffusion Models
(2020)

$p(z) = \mathcal{N}$ $p(z)$ $p(z)$ arbitrary



Flow-Matching
(2023)

$p(z)$ arbitrary

[cs.CV] 12 Mar 2026

Latent diffusion models for parameterization and data assimilation of facies-based geomodels

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Abstract

Geological parameterization entails the representation of a geomodel using a small set of latent variables and a mapping from these variables to grid-block properties such as porosity and permeability. Parameterization is useful for data assimilation (history matching), as it maintains geological realism while reducing the number of variables to be determined. Diffusion models are a new class of generative deep-learning procedures that have been shown to outperform previous methods, such as generative adversarial networks, for image generation tasks. Diffusion models are trained to "denoise", which enables them to generate new geological realizations from input fields characterized by random noise. Latent diffusion models, which are the specific variant considered in this study,

- Di Federico, Guido, and Louis J. Durlofsky. "Latent diffusion models for parameterization of facies-based geomodels and their use in data assimilation." *Computers & Geosciences* 194 (2025): 105755.
- Xu, Yixian, Shengjie Luo, Liwei Wang, Di He, and Chang Liu. "Diagnosing and improving diffusion models by estimating the optimal loss value." *arXiv preprint arXiv:2506.13763* (2025).

Outline

1. Kalman filter in 1D
2. Derivation and example

General Definition of Filtering

Filtering (also known as data assimilation or state estimation) is the process of obtaining the best statistical estimate of a natural system from partial observations of the true signal from nature.

- ▶ $\text{Model} + \text{Data} = \text{Data Assimilation}.$

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- ▶ Model + Data = Data Assimilation.
- ▶ Model with bias and uncertainty + Data with noise = Data Assimilation.

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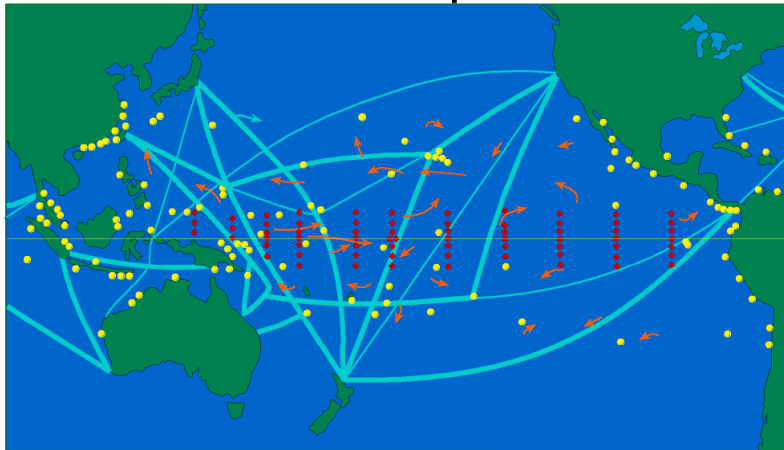
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e.g.,

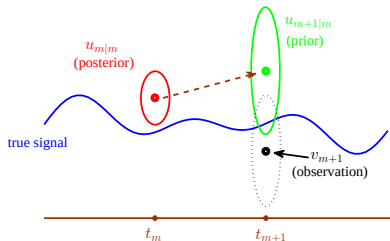
Model	(30% error + 50% uncertainty)
+ Data	(30% error + 50% uncertainty)
<u>data assimilation</u> →	Estimation (10% error + 20% uncertainty)

TOGA In Situ Ocean Observing System Global Tropics

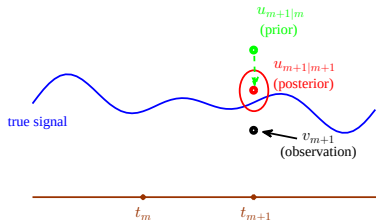


Filtering: A Two-Step Procedure

1. Prediction (Forecast)



2. Analysis (Filtering)

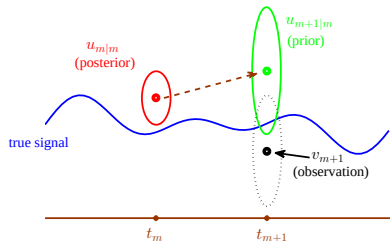


Two steps at each time instant $t_m = m\Delta t$.

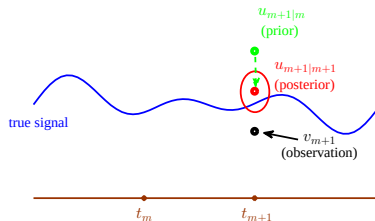
Step 1. A statistical prediction of a probability distribution $u_{m+1|m}$ starting from the initial value $u_{m|m}$ using the **given dynamical model**.

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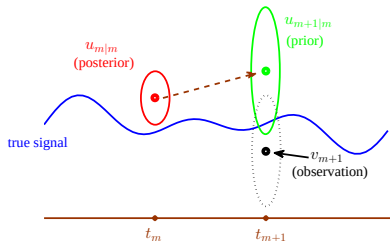


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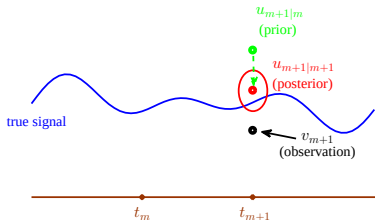
- Step 1.** A statistical prediction of a probability distribution $u_{m+1|m}$ starting from the initial value $u_{m|m}$ using the **given dynamical model**.
- Step 2.** $u_{m+1|m}$ is corrected on the basis of the statistical input of **noisy observation** v_{m+1} , which results in $u_{m+1|m+1}$.

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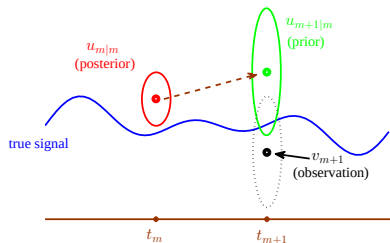
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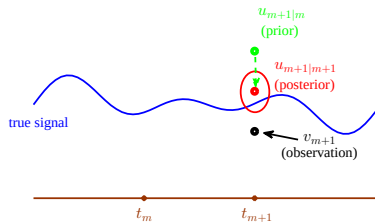
For linear system with Gaussian noise, the above procedure is known as the *Kalman filter*.

Filtering: 1-D Complex Case

1. Prediction (Forecast)



2. Analysis (Filtering)



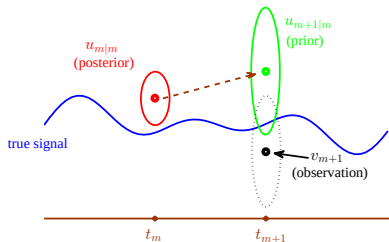
Let $u_m \in \mathbb{C}$ be a complex random variable whose dynamics is given by the following,

$$u_{m+1} = Fu_m + \mathcal{F}_{m+1} + \sigma_{m+1},$$

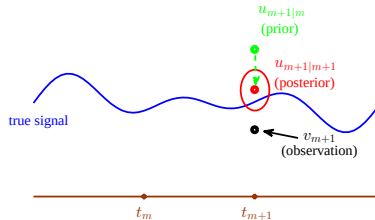
- ▶ σ_{m+1} is a complex Gaussian noise with $\sigma_{m+1} = (\sigma_{1,m+1} + i\sigma_{2,m+1})/\sqrt{2}$ and it has zero mean and variance $r = \langle \sigma_{m+1} \sigma_{m+1}^* \rangle = \frac{1}{2} \sum_{j=1}^2 \langle \sigma_{j,m+1}^2 \rangle$.
- ▶ F is a complex number known as the forward operator.
- ▶ $\mathcal{F}$ is an external forcing which can vary in time.

Filtering: 1-D Complex Case

1. Prediction (Forecast)



2. Analysis (Filtering)



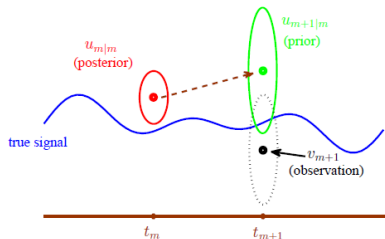
The goal of the Kalman filter is to estimate the unknown true state u_{m+1} , given noisy observations

$$v_{m+1} = g u_{m+1} + \sigma_{m+1}^o,$$

- ▶ g is a linear observation operator.
- ▶ $\sigma_m^o \in \mathbb{C}$ is an unbiased Gaussian noise with variance $r^o = \langle \sigma_m^o (\sigma_m^o)^* \rangle$.

Filtering: 1-D Complex Case

1. Prediction (Forecast)



Model:	$u_{m+1} = Fu_m + \mathcal{F}_{m+1} + \sigma_{m+1},$
Obs:	$v_{m+1} = gu_{m+1} + \sigma_{m+1}^o.$

Assume the filter model is perfectly specified.

An estimate of the true state prior to knowledge of the observation at time t_{m+1} , which is known as the **prior state or forecast state**, is given by

$$u_{m+1|m} = Fu_{m|m} + \mathcal{F}_{m+1} + \sigma_{m+1}.$$

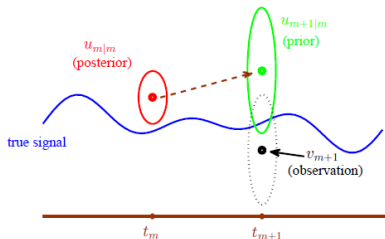
This prior estimate can be represented with a probability density

$$p(u_{m+1}) \sim \mathcal{N}(\bar{u}_{m+1|m}, r_{m+1|m}),$$

which accounts only for the earlier observations up to time t_m .

Filtering: 1-D Complex Case

1. Prediction (Forecast)



$$\text{Model: } u_{m+1} = Fu_m + \mathcal{F}_{m+1} + \sigma_{m+1},$$

$$\text{Obs: } v_{m+1} = gu_{m+1} + \sigma_{m+1}^o.$$

The prior mean and prior variance

$$\bar{u}_{m+1|m} \equiv \langle u_{m+1|m} \rangle,$$

$$r_{m+1|m} \equiv \langle (u_{m+1} - \bar{u}_{m+1|m})(u_{m+1} - \bar{u}_{m+1|m})^* \rangle,$$

are solved via

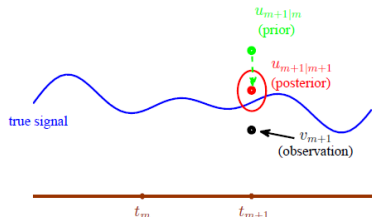
$$\bar{u}_{m+1|m} = F\bar{u}_{m|m} + \mathcal{F}_{m+1},$$

$$r_{m+1|m} = Fr_{m|m}F^* + r,$$

with $r_{m|m} = \langle (u_m - \bar{u}_{m|m})(u_m - \bar{u}_{m|m})^* \rangle$. Note that in order to solve the prior distribution $p(u_{m+1|m})$, the posterior information in the previous step $\bar{u}_{m|m}, r_{m|m}$ has been used.

Filtering: 1-D Complex Case

2. Analysis (Filtering)



$$\text{Model: } u_{m+1} = Fu_m + \mathcal{F}_{m+1} + \sigma_{m+1},$$

$$\text{Obs: } v_{m+1} = gu_{m+1} + \sigma_{m+1}^o.$$

The posterior state (or the filtered state) combines the prior information $p(u_{m+1}|m)$ with the observation v_{m+1} at t_{m+1} .

This estimate is given in the probabilistic sense by the Bayesian update through maximizing the following conditional density,

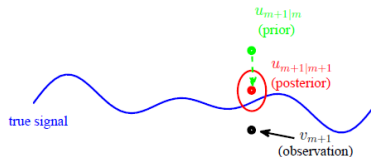
$$p(u_{m+1}|v_{m+1}) \sim p(u_{m+1})p(v_{m+1}|u_{m+1}) = e^{-\frac{1}{2}J(u_{m+1})},$$

which is equivalent to minimizing

$$J(u) = \frac{(u - \bar{u}_{m+1|m})^*(u - \bar{u}_{m+1|m})}{r_{m+1|m}} + \frac{(v_{m+1} - gu)^*(v_{m+1} - gu)}{r^o}.$$

Filtering: 1-D Complex Case

2. Analysis (Filtering)



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The value of u at which $J(u)$ attains its minimum is the estimate for the mean

$$\bar{u}_{m+1|m+1} = (1 - K_{m+1}g)\bar{u}_{m+1|m} + K_{m+1}v_{m+1},$$

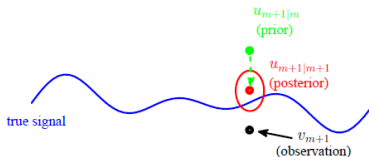
where

$$K_{m+1} = \frac{gr_{m+1|m}}{r^o + g^2r_{m+1|m}}$$

is the **Kalman gain**. Note that $0 \leq K_{m+1}g \leq 1$.

Filtering: 1-D Complex Case

2. Analysis (Filtering)



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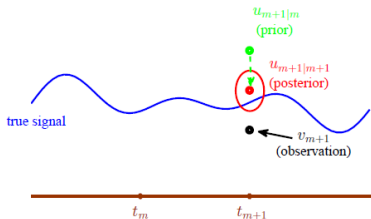
$$K_{m+1} = \frac{gr_{m+1|m}}{r^o + g^2r_{m+1|m}}$$

is the **Kalman gain**. Note that $0 \leq K_{m+1}g \leq 1$.

The filter fully weighs to the model or prior forecast when $K_{m+1}g = 0$ and fully weighs to the observation when $K_{m+1}g = 1$. Such weights depend on the ratio of the uncertainty (reflected by the noise) in the observations and the model.

Filtering: 1-D Complex Case

2. Analysis (Filtering)



$$\text{Model: } u_{m+1} = Fu_m + \mathcal{F}_{m+1} + \sigma_{m+1},$$

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Posterior mean:

$$\bar{u}_{m+1|m+1} = (1 - K_{m+1}g)\bar{u}_{m+1|m} + K_{m+1}v_{m+1}.$$

Finally, the posterior variance is calculated via the following

$$\begin{aligned} u_{m+1} - \bar{u}_{m+1|m+1} &= u_{m+1} - \bar{u}_{m+1|m} - K_{m+1}(v_{m+1} - gu_{m+1} - g(\bar{u}_{m+1} - u_{m+1})) \\ e_{m+1|m+1} &= (1 - K_{m+1}g)e_{m+1|m} - K_{m+1}\sigma_{m+1}^o. \end{aligned}$$

These result in the expression of the posterior variance

$$r_{m+1|m+1} = (1 - K_{m+1}g)r_{m+1|m}.$$

Note that the above Kalman filter is designed for **linear system with Gaussian noise**. In practice, different generalizations of the Kalman filter such as the ensemble Kalman filter, particle filter and blended filtering techniques are applied to nonlinear and non-Gaussian systems.

Example.

Consider a complex scalar forced OU process,

$$\frac{du}{dt} = (-\gamma + i\omega_0)u + f_0 + f_1 e^{i\omega_1 t} + \sigma \dot{W},$$

- ▶ γ and ω_0 are the damping and oscillation frequency.
- ▶ f_0 and $f_1 e^{i\omega_1 t}$ are a constant and time-periodic large-scale forcing.
- ▶ $\sigma \dot{W}$ is stochastic noise.

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- ▶ $\sigma \dot{W}$ is stochastic noise.

Despite the simplicity of this model, it can be used to mimic some climate physics, e.g.,

- ▶ The deterministic forcing $f_0 + f_1 e^{i\omega_1 t}$ can be regarded as the annual cycle.
- ▶ ω_0 can be treated as internal oscillation which may occur in the intraseasonal time scale.
- ▶ The damping term γ measures the system memory and the stochastic term represents the input to the system from small or unresolved scales.
- ▶ The model can also be regarded as one Fourier mode of complex spatially-extended systems.

The simple test model:

$$\frac{du}{dt} = (-\gamma + i\omega_0)u + f_0 + f_1 e^{i\omega_1 t} + \sigma \dot{W}.$$

How to put this model into the framework of Kalman filter?

Model: $u_{m+1} = Fu_m + \mathcal{F}_{m+1} + \sigma_{m+1},$

Obs: $v_{m+1} = gu_{m+1} + \sigma_{m+1}^o.$

$$\frac{du}{dt} = (-\gamma + i\omega_0)u + f_0 + f_1 e^{i\omega_1 t} + \sigma \dot{W} \quad \Longrightarrow \quad u_{m+1} = F u_m + \mathcal{F}_{m+1} + \sigma_{m+1}$$

The exact solution can be written down explicitly,

$$u(t) = u(t_0) e^{(-\gamma + i\omega_0)(t-t_0)} + \frac{f_0}{\gamma - i\omega_0} (1 - e^{(-\gamma + i\omega_0)(t-t_0)}) \\ + \frac{f_1 e^{i\omega_1 t}}{\gamma + i(-\omega_0 + \omega_1)} \left(1 - e^{-(\gamma + i\omega_1 - i\omega_0)(t-t_0)} \right) + \sigma \int_{t_0}^t e^{(-\gamma + i\omega_0)(t-s)} dW(s),$$

which provides the analytical forms of the time evolution of the forecast mean $\bar{u}(t)$ and the forecast variance $r(t)$,

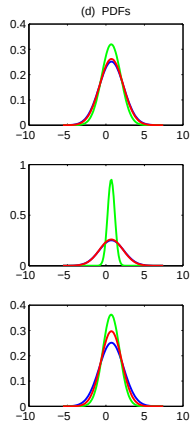
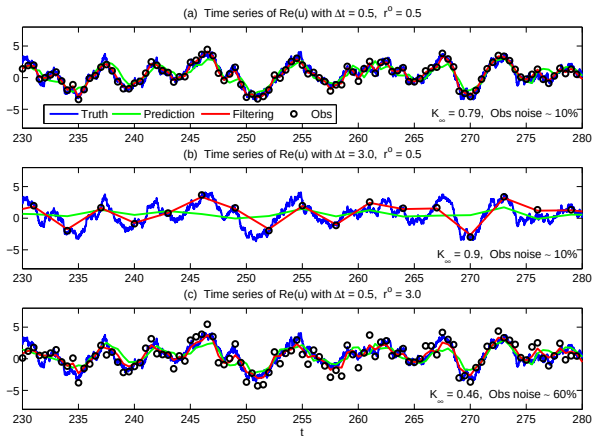
$$\bar{u}(t) = \bar{u}(t_0) e^{(-\gamma + i\omega_0)(t-t_0)} + \frac{f_0}{\gamma - i\omega_0} (1 - e^{(-\gamma + i\omega_0)(t-t_0)}) \\ + \frac{f_1 e^{i\omega_1 t}}{\gamma + i(-\omega_0 + \omega_1)} \left(1 - e^{-(\gamma + i\omega_1 - i\omega_0)(t-t_0)} \right), \\ r(t) = r(t_0) e^{-2\gamma(t-t_0)} + \frac{\sigma^2}{2\gamma} (1 - e^{-2\gamma(t-t_0)}).$$

The test model

$$\frac{du}{dt} = (-\gamma + i\omega_0)u + f_0 + f_1 e^{i\omega_1 t} + \sigma \dot{W}$$

with parameters

$$\gamma = 0.4, \quad \omega_0 = 1, \quad f_0 = 2, \quad f_1 = 0, \quad \omega_1 = 0, \quad \sigma = 1, \quad g = 1.$$



The N -dimensional Kalman filter

Our goal is to obtain the posterior distribution $p(\vec{u}|\vec{v}_{m+1})$ through the Bayesian formula assuming a knowledge of the prior distribution $p(\vec{u})$ of a true signal $\vec{u}_{m+1} \in \mathbb{R}^N$ and the observed value $\vec{v}_{m+1} \in \mathbb{R}^M$.

The model is

$$\vec{u}_{m+1|m} = F\vec{u}_{m|m} + \vec{\sigma}_{m+1}, \quad R = \langle \vec{\sigma}_m \otimes \vec{\sigma}_m^T \rangle.$$

The observational process is

$$\vec{v}_{m+1} = G\vec{u}_{m+1} + \vec{\sigma}_{m+1}^o, \quad R^o = \langle \vec{\sigma}_m^o \otimes (\vec{\sigma}_m^o)^T \rangle.$$

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The posterior mean and posterior covariance are given by

$$\begin{aligned} \vec{u}_{m+1|m+1} &= \vec{u}_{m+1|m} + K_{m+1}(\vec{v}_{m+1} - G\vec{u}_{m+1|m}), \\ R_{m+1|m+1} &= (I - K_{m+1}G)R_{m+1|m} \end{aligned}$$

where the Kalman gain is

$$K_{m+1} = R_{m+1|m}G^T(GR_{m+1|m}G^T + R^o)^{-1}.$$

Time Evolution of the Posterior Statistics

In general, the Kalman filter system is given by

$$\mathbf{u}_I(t_{i+1}) = \mathbf{G}\mathbf{u}_{II}(t_{i+1}) + \mathbf{B}_1\varepsilon_1,$$

$$\mathbf{u}_{II}(t_{i+1}) = \mathbf{a}_0 + \mathbf{a}_1\mathbf{u}_{II}(t_i) + \mathbf{b}_2\varepsilon_2,$$

Define $\mathbf{A}_0 = \mathbf{G}\mathbf{a}_0$, $\mathbf{A}_1 = \mathbf{G}\mathbf{a}_1$ and $\mathbf{B}_2 = \mathbf{G}\mathbf{b}_2$. Then the above system becomes

$$\mathbf{u}_I(t_{i+1}) = \mathbf{A}_0 + \mathbf{A}_1\mathbf{u}_{II}(t_i) + \mathbf{B}_1\varepsilon_1 + \mathbf{B}_2\varepsilon_2,$$

$$\mathbf{u}_{II}(t_{i+1}) = \mathbf{a}_0 + \mathbf{a}_1\mathbf{u}_{II}(t_i) + \mathbf{b}_2\varepsilon_2,$$

The time evolutions of the posterior mean and posterior covariance are given by

$$\begin{aligned}\boldsymbol{\mu}(t_{i+1}) &= \mathbf{a}_0 + \mathbf{a}_1\boldsymbol{\mu}(t_i) + (\mathbf{b}_2\mathbf{B}_2^* + \mathbf{a}_1\mathbf{R}(t_i)\mathbf{A}_1^*) \times \\ &\quad (\mathbf{B}_1\mathbf{B}_1^* + \mathbf{B}_2\mathbf{B}_2^* + \mathbf{A}_1\mathbf{R}(t_i)\mathbf{A}_1^*)^{-1} (\mathbf{u}_I(t_{i+1}) - \mathbf{A}_0 - \mathbf{A}_1\boldsymbol{\mu}(t_i)), \\ \mathbf{R}(t_{i+1}) &= \mathbf{a}_1\mathbf{R}(t_i)\mathbf{a}_1^* + \mathbf{b}_2\mathbf{b}_2^* - (\mathbf{b}_2\mathbf{B}_2^* + \mathbf{a}_1\mathbf{R}(t_i)\mathbf{A}_1^*) \times \\ &\quad (\mathbf{B}_1\mathbf{B}_1^* + \mathbf{B}_2\mathbf{B}_2^* + \mathbf{A}_1\mathbf{R}(t_i)\mathbf{A}_1^*)^{-1} (\mathbf{b}_2\mathbf{B}_2^* + \mathbf{a}_1\mathbf{R}(t_i)\mathbf{A}_1^*)^*.\end{aligned}$$

The equation of $\mathbf{R}$ is called the Riccati equation.

Thank you!

References and acknowledgements

1. Eugenia Kalnay. Atmospheric modeling, data assimilation and predictability. Cambridge university press, 2003.
2. Kody Law, Andrew Stuart, and Kostas Zygalakis. Data assimilation. Cham, Switzerland: Springer, 2015.
3. Andrew J Majda, and John Harlim. Filtering complex turbulent systems. Cambridge University Press, 2012.
4. We gratefully acknowledge that part of these lecture notes is adapted from Nan Chen's original materials and other online resources.